



MOBILE COMMUNICATION SYSTEM

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LECTURE

__1__

Introduction to **WIRELESS COMMUNICATIONS**



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Fourth Level – Computer Science

1st Semester

Code	Course title	Contact hours		Credit hours	Mark Distribution						Exam Time	Prerequisite
		Lecture	Practical		Final Exam	Practical Exam	Mid-Term Exam	Course Activities	Oral Exam	Total		
CSD441	Mobile Communication Systems	2	2	3	50	20	10	15	5	100	2	CSD338



MOBILE COMMUNICATION SYSTEM



WIRELESS
TECHNOLOGY



NETWORK
CONNECTIVITY



DATA
TRANSMISSION



SIGNAL
PROCESSING



MOBILE
NETWORKS



GLOBAL
COMMUNICATION



LECTURE CONTENT

- ✓ Rising Need for Wireless Communication
- ✓ Wireless Network Examples
- ✓ Mobile Technologies from 1G to 5G
- ✓ Mobile generation: Services and Statistics
- ✓ WiFi Standards from WiFi 1 to WiFi 5
- ✓ WiFi 6 & 5G Complement Each Other
- ✓ Internet of Things
- ✓ Mobile Connectivity Statistics
- ✓ Wireless Characteristics
- ✓ Wireless Types



Rising Need for Wireless Communication

Example from the past: people were less connected than today

In the Past (around 2005)



- At public events, only a **few** mobile **devices** were visible.



- Some people used **digital cameras** or **camcorders** instead of connected smartphones.



- Wireless access such as Wi-Fi and 3G was **limited**, slower, and more expensive.



- Most people were not always online, so the overall **demand** for wireless communication was **lower**.



Key idea: Because fewer people and fewer devices were connected all the time, wireless traffic demand was much smaller than it is today.



Typical public event scene – around 2005

Rising Need for Wireless Communication

Now: people are connected all the time



Now (2025)

- Almost everyone in a crowd is holding a **smartphone**.
- These devices depend heavily on **wireless** connectivity such as **4G, 5G, Wi-Fi**, and **Bluetooth**.
- People use wireless networks to take **photos**, upload to **social media**, **live stream**, **message**, and **share** instantly.
- As a result, **mobile internet traffic** has grown dramatically because more users are connected and each user generates more data.



2025



Key implication

Wireless networks now need much higher **speed**, **capacity**, **reliability**, and **coverage** to support modern digital life.



More Users



More Data



Always Connected



Higher Network Demand

Wireless Networks

Definition and Basic Idea



A wireless network is a communication network that connects devices without physical cables by using electromagnetic waves such as radio, microwave, or infrared signals.



1. How it works

- 1 Devices send and receive data through wireless signals.
- 2 An access point or base station manages communication.
- 3 The network allows users to share information and access the internet.



2. Main features

- No physical cables
- Mobility and flexibility
- Easy installation
- Supports many devices



3. Common types

-  Wi-Fi (WLAN)
-  Cellular networks
-  Bluetooth
-  Satellite wireless links



4. Examples of use

- Home and office internet
- Mobile communication
- Smart homes and IoT
- Campus and public hotspots



Wireless networks make communication easier by allowing devices to connect anytime and anywhere within coverage range.



Why Wireless Connectivity Dominates Today's World

1 / 2

Part 1: Mobility, Device Growth, and High-Speed Access

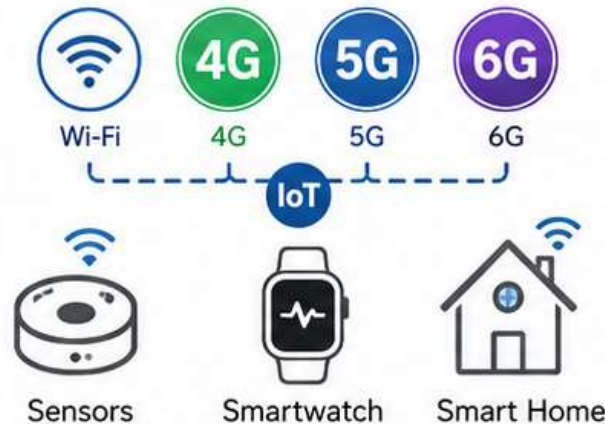
1 Mobility and Convenience

- ✓ Connect anywhere, anytime
- ✓ No cables required
- ✓ Ideal for phones, laptops, tablets, and wearables



2 Explosion of Mobile Devices

- ✓ Billions of devices depend on wireless links
- ✓ Wi-Fi, LTE/4G, 5G, and emerging 6G
- ✓ IoT devices such as sensors, appliances, and smartwatches are wireless



3 High-Speed Wireless Technologies

- ✓ Wi-Fi 6/6E/7 and 5G offer faster speeds
- ✓ Lower latency supports real-time communication
- ✓ In some cases, wireless can rival wired performance



TAKEAWAY: Wireless connectivity dominates because it is **mobile**, **widely adopted**, and **increasingly fast**.



Why Wireless Connectivity Dominates Today's World

Part 2: Cost, Applications, and Industry Adoption

Slide 2 of 2

4 Cost-Effective Infrastructure



Faster and cheaper to deploy than laying cables



Useful in rural, remote, and dense urban areas



Practical where wiring is expensive or difficult



5 Support for Modern Applications



Video streaming and cloud services



Real-time communication and cloud gaming



AR/VR, IoT, and smart city applications



6 Enterprise and Industrial Adoption



Used in logistics and remote work



Supports business operations and automation



Enables sensor-based monitoring in industry



Key Message

Wireless networks dominate modern life because they are flexible, scalable, affordable, and essential for consumer and industrial applications.



Wireless Network Examples

Wireless communication is no longer limited to phones — it powers entire ecosystems.

1 Smart Home



Wi-Fi connects lights, TVs, cameras, and home appliances.

2 Wireless Sensors



Sensors send data for agriculture, environment, and industry.

3 Cellular Networks



4G/5G towers connect mobile users and smart devices.

4 Wearables



Smart watches and health devices use Bluetooth and Wi-Fi.

5 Drones



Drones use wireless links for control, video, and navigation.

6 Wireless Data Centers



Servers and edge systems exchange data in connected infrastructures.

7 AR / VR



Wireless links support immersive virtual and augmented reality experiences.

8 Connected Vehicles



Vehicles communicate for safety, traffic management, and smart transport.



Wireless networks support communication, sensing, control, and smart services across many applications.

Every New Wireless Mobile Generation Requirements

— Each generation brings major improvements to meet the growing needs of users and new technologies. —



Speed

- ✓ Higher peak data rates
- ✓ Faster downloads & uploads
- ✓ Better experience for streaming, gaming, AR/VR



Device Quality

- ✓ Smaller size & lighter weight
- ✓ Longer battery life
- ✓ Better display & sound
- ✓ Stronger & smarter hardware



More Functions

- ✓ More services & applications
- ✓ Internet, video, messaging
- ✓ IoT, cloud & smart services
- ✓ AI, AR/VR, smart devices



Transmission Quality

- ✓ Better signal quality
- ✓ Lower latency & jitter
- ✓ Wider coverage
- ✓ More reliable connections



Security

- ✓ Stronger encryption
- ✓ Protect user privacy
- ✓ Secure authentication
- ✓ Defense against cyber threats



2G



3G



3.5G / HSPA



4G



4G LTE



5G



5G Advanced



Future (6G and Beyond)

NEWER REQUIREMENTS (2025 AND BEYOND)



Ultra High Speed

Multi-Gbps speeds for seamless experiences.



Ultra Low Latency

Real-time response for mission-critical apps, gaming, and AR/VR.



High Reliability

Consistent connections with minimal downtime and interruptions.



Energy Efficiency

Lower power consumption for devices and network sustainability.



Massive Connectivity

Connect billions of IoT devices and smart everything.



Security & Privacy

End-to-end security, privacy-by-design and data protection.

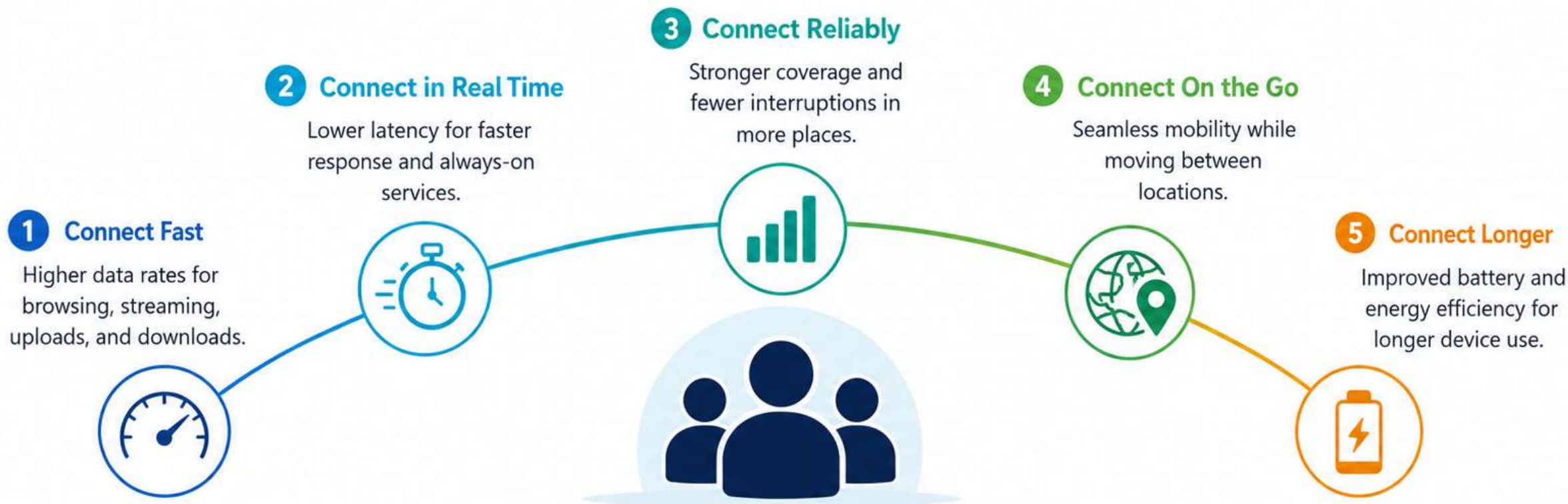


Better Coverage

Seamless connectivity everywhere – indoors, outdoors & remote areas.

Every New Wireless Mobile Generation Delivers More

Each new mobile generation improves user experience through speed, latency, reliability, mobility, and energy efficiency.



Delivering Rich Mobile Broadband Experiences



Main technical goals: higher speed, lower latency, better reliability, seamless mobility, and better energy efficiency.

Evolution of Mobile Generations (1G to 5G up to 2025)

— How mobile communication evolved over time —



1G

Analog voice

- Voice only
- No data services
- Early analog cellular systems



2G

Digital voice & SMS

- Clearer digital calls
- SMS text messaging
- Basic mobile data
- Typical speeds below 0.5 Mbps



3G

Mobile internet

- Faster wireless data
- Web browsing and email
- Video calling
- Early mobile apps and games



4G

Mobile broadband

- Much faster than 3G
- HD video streaming
- Better app performance
- Lower latency and smoother gaming



5G

Gigabit connectivity

- Very high data rates
- Ultra-low latency
- Massive IoT support
- Better reliability and capacity
- Supports AI, cloud services, AR/VR and smart cities

1980s

AMPS, NMT, TACS

1990s

GSM, D-AMPS,
PDC, GPRS, EDGE

2000s

UMTS, HSPA,
CDMA2000

2010s

LTE, LTE-Advanced

2020–2025

5G NR, 5G Advanced

💡 Commercial 6G is not yet deployed in 2025; research is ongoing.

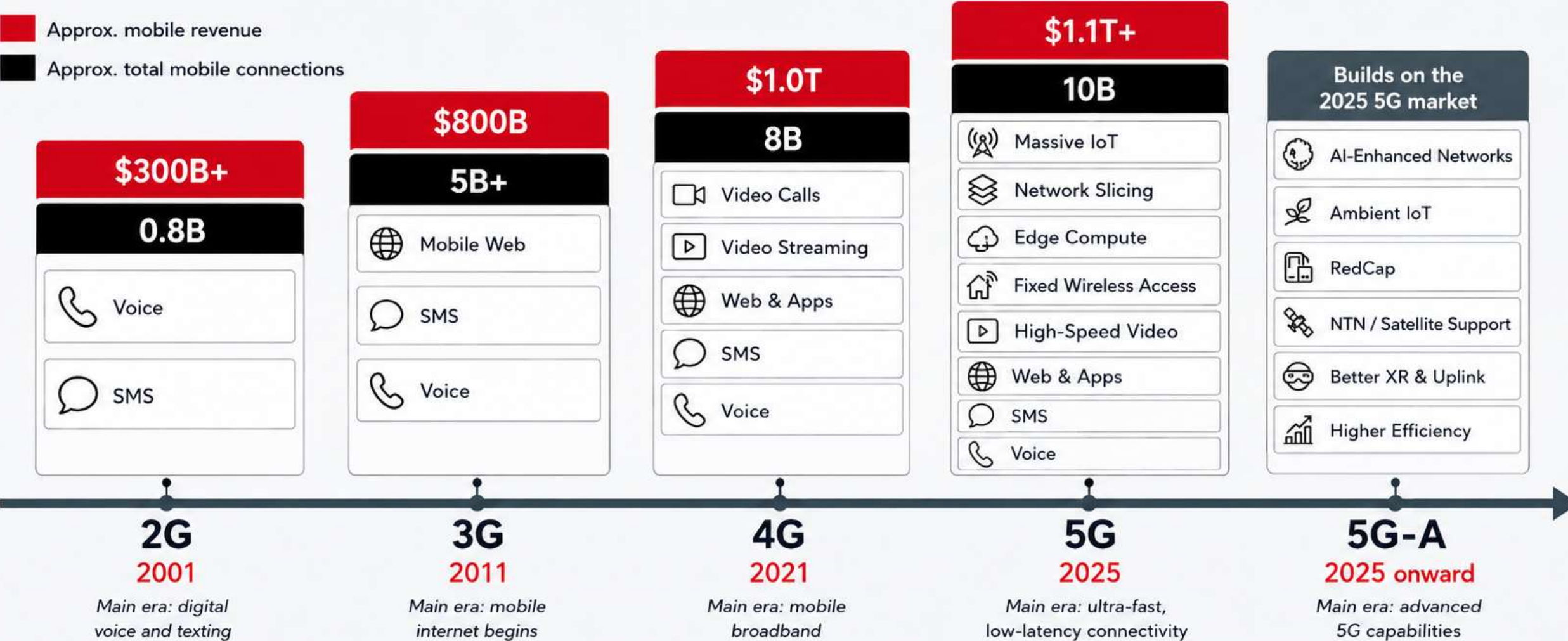


Each generation improved speed, capacity, mobility, and user experience.



5G-Advanced: The Next Step in Cellular Technology

Simplified global view updated through 2025

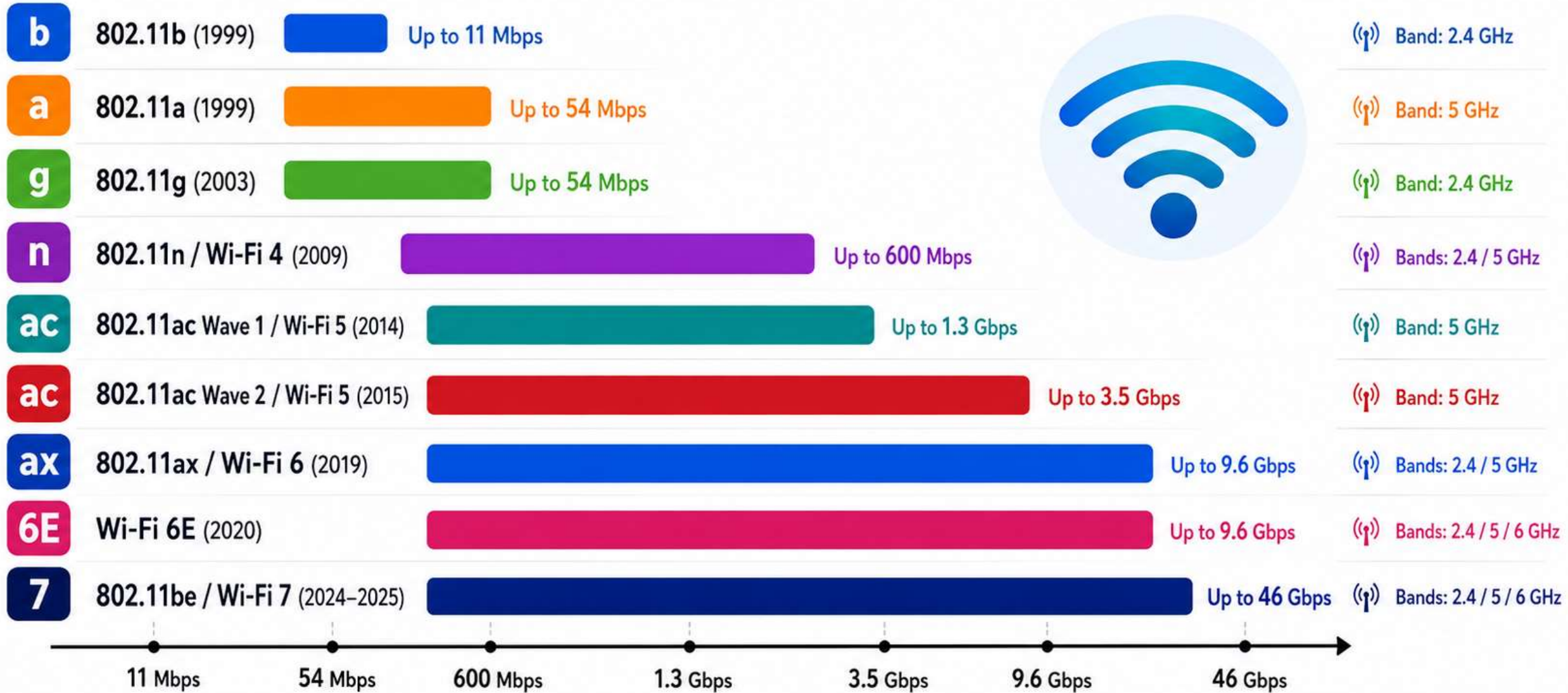


Source: simplified educational summary based on GSMA industry outlook and 2025 mobile market estimates.

 6G is not commercially deployed in 2025. Research is ongoing.

Wi-Fi Standards Evolution (1999–2025)

Mainstream consumer Wi-Fi generations from 802.11a/b/g to Wi-Fi 7



i Focuses on widely used mainstream Wi-Fi generations.

Speeds shown are maximum theoretical rates.

Wi-Fi 6 and 5G Complement Each Other

They are not competitors — they work together to provide complete wireless connectivity.



What is each optimized for?



5G

- ✓ Licensed spectrum
- ✓ Wide-area coverage
- ✓ Seamless mobility and handover
- ✓ Supports eMBB, URLLC, and mMTC
- ✓ Carrier-grade security and QoS
- ✓ Best for transportation, public networks, and massive IoT



Wi-Fi 6 (802.11ax)

- ✓ Unlicensed spectrum
- ✓ High capacity indoor wireless LAN
- ✓ Lower-cost local deployment
- ✓ Uses OFDMA, MU-MIMO, BSS Coloring, and TWT
- ✓ Best for dense indoor spaces
- ✓ Ideal for homes, offices, retail, campuses, and AR/VR



Transportation



Outdoor Coverage



AR/VR



Massive IoT



Indoor Networks



Retail & Campus



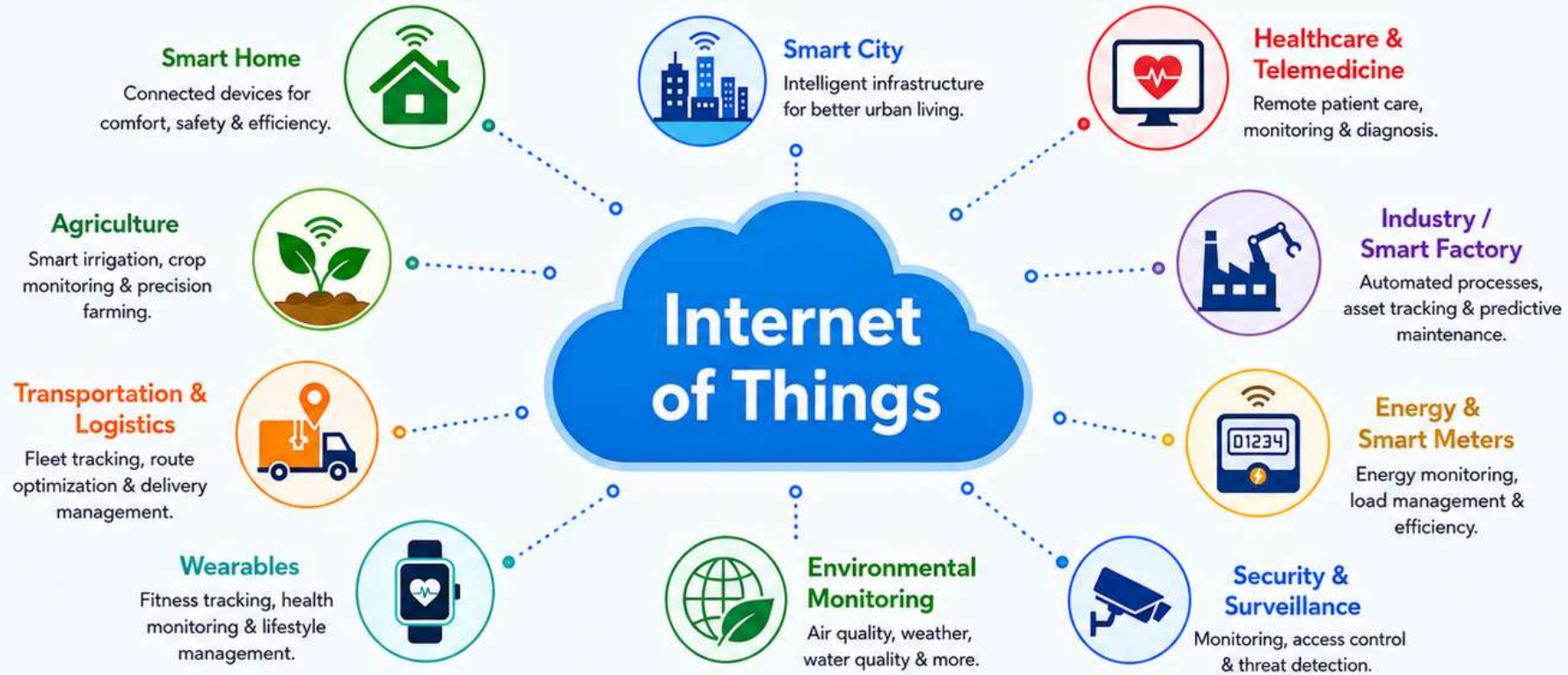
eMBB = enhanced Mobile Broadband
URLLC = ultra-reliable low-latency communication
mMTC = massive Machine-Type Communications



Key idea: 5G provides wide-area mobile connectivity, while Wi-Fi 6 provides high-capacity indoor connectivity. Together, they create a complete wireless experience.

Internet of Things (IoT)

Connecting everyday objects to the internet so they can collect, share, and use data.



What is IoT?

IoT refers to physical devices such as sensors, appliances, vehicles, and machines that are connected to the internet. These devices can sense data, communicate with each other, and help people make better decisions.

How IoT works

1 Sense

devices collect data using sensors.



2 Connect

data is sent through a network.



3 Analyze & Act

software analyzes the data and triggers actions.

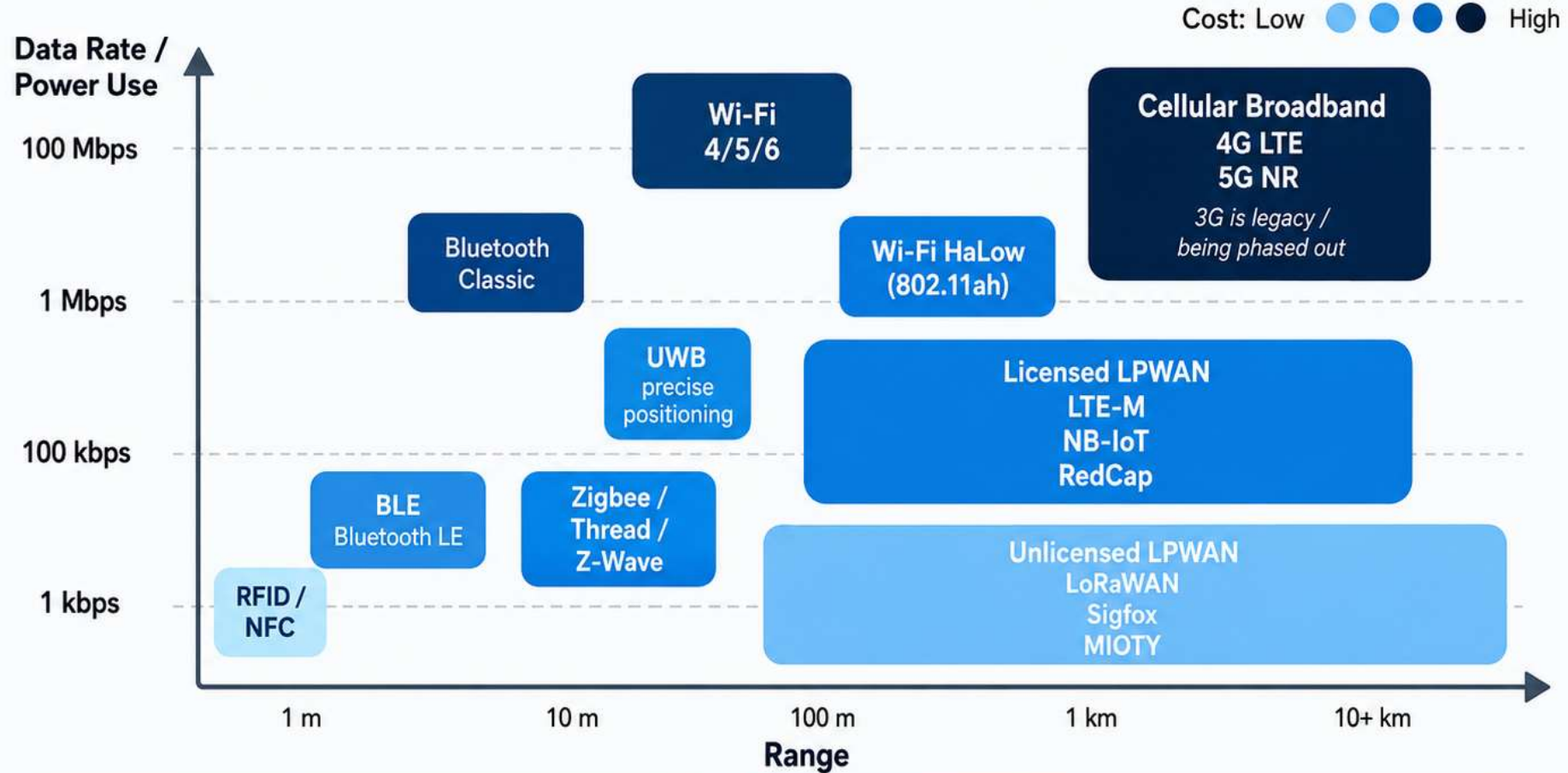


Examples: smart lights, fitness bands, connected cars, medical monitors, and smart irrigation systems.



IoT Connectivity Technologies (Updated to 2025)

A simple comparison of common wireless technologies used in Internet of Things systems.



How to read this chart

- Higher on the chart = higher data rate, but often more power consumption.
- Farther right = longer communication range.
- Different technologies are chosen based on battery life, range, data volume, and cost.



Typical use cases

- **RFID / NFC:** identification and access cards
- **BLE / Zigbee / Thread:** smart home sensors and wearables
- **LPWAN:** smart metering, agriculture, remote monitoring
- **Wi-Fi / Cellular:** cameras, gateways, and high-data IoT devices

Wireless Characteristics

Channel Gain



DEFINITION

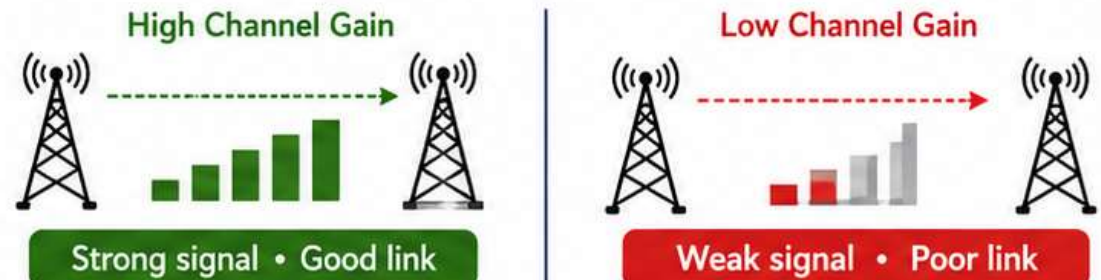
Channel gain is the net signal strength or power received after a signal passes through the wireless channel. It reflects the combined effect of distance, path loss, antennas, fading, and the environment.



KEY IDEAS

- 1 Channel gain measures channel strength / quality.
- 2 Higher channel gain → stronger received signal.
- 3 Lower channel gain → more attenuation and weaker reception.
- 4 Channel gain is affected by distance, obstacles, fading, and antenna characteristics.

CHANNEL GAIN COMPARISON



In short: higher channel gain means better signal reception and more reliable communication.

Wireless Characteristics

Channel Gain



Definition

Channel gain describes how much of the transmitted signal is effectively received after the wireless channel affects it. It reflects the combined impact of distance, obstacles, fading, and motion.

Factors Influencing Channel Gain

1



Path Loss

Signal attenuation due to the distance between transmitter and receiver.

2



Shadowing

Large obstacles such as buildings or trees cause additional signal loss.

3



Multipath Fading

Reflections, refractions, and diffraction create constructive and destructive interference.

4



Doppler Shift

Relative motion between transmitter and receiver changes the received frequency and affects the signal.



Key Ideas

- Higher channel gain → stronger received signal
- Lower channel gain → more attenuation and weaker reception
- Channel gain changes with distance, obstacles, reflections, and motion



Channel Gain vs. Distance

Channel gain
(log scale)

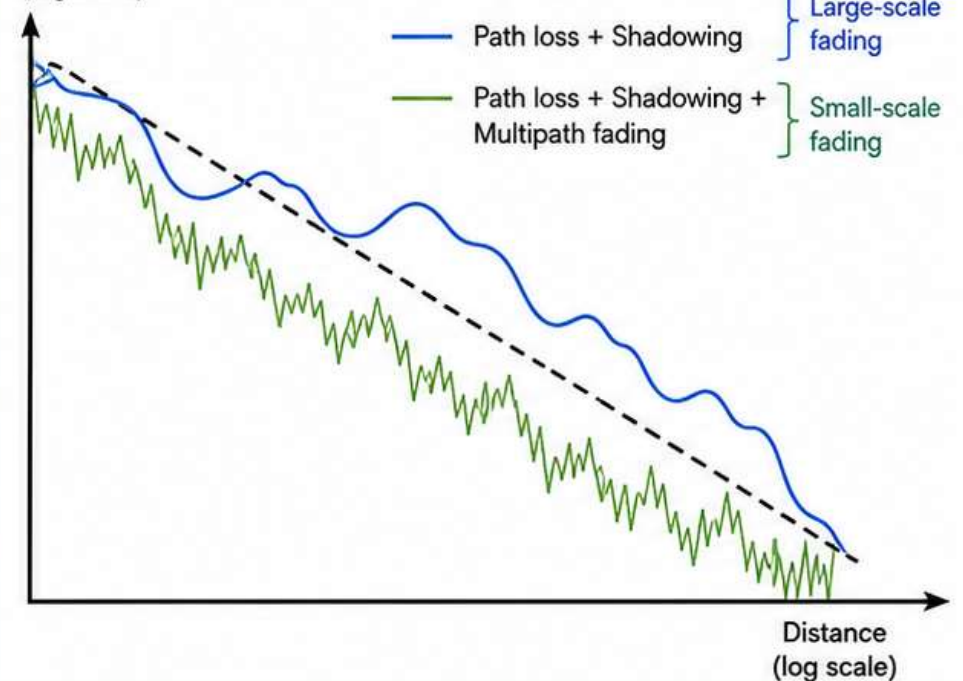
--- Path loss

— Path loss + Shadowing

— Path loss + Shadowing +
Multipath fading

} Large-scale
fading

} Small-scale
fading



In short: channel gain measures the strength and quality of the wireless channel seen by the receiver.

Wireless Characteristics

Channel Gain



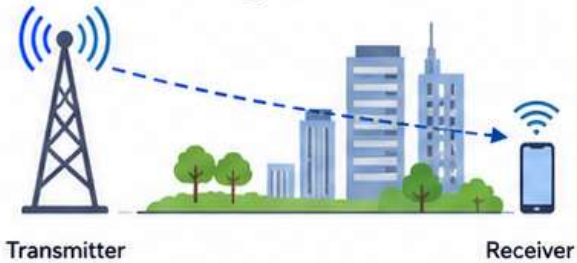
DEFINITION

Channel gain describes how strongly a transmitted signal is received after passing through the wireless channel. It affects the quality and strength of the received signal.

1) TYPES OF CHANNEL GAIN

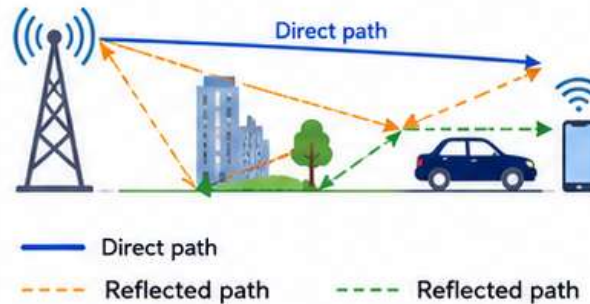
Large-Scale Gain

Represents slow changes in signal level caused mainly by path loss and shadowing over distance.



Small-Scale Gain

Represents rapid signal fluctuations caused by multipath propagation and motion.



2) APPLICATIONS

Affects
Signal-to-Noise
Ratio (SNR)



Influences
Data Rate



Impacts
Error Performance



Used in
Link Adaptation



Important for
Beamforming



Used in
Power Control



3) PRACTICAL MEANING



Stronger
Channel Gain



Better
Received Signal



Higher
Quality Link



Weaker
Channel Gain



Lower
Received Signal



Poorer
Link Quality



KEY TAKEAWAY: Channel gain is important because it directly affects signal quality, data rate, reliability, and wireless system design.

Wireless Characteristics

Key Characteristics of Wireless Environments

1 Time-Varying Environment



- Channel conditions change over time.
- User mobility affects signal strength and quality.
- Interference and errors vary with location and time.

2 Broadcast Nature of the Channel



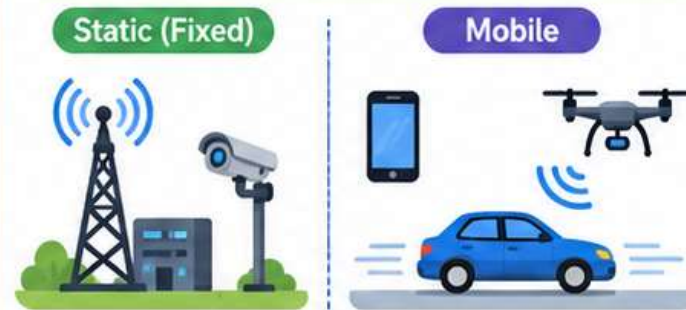
- Wireless signals spread through the air.
- Many users share the same medium.
- Multiple access methods are needed to avoid collisions.

3 Less-Secure Environment



- Signals can be intercepted more easily than wired links.
- Security is important because the medium is open.
- Encryption and authentication help protect data.

4 Wireless Nodes Can Be Static or Mobile



- Some wireless nodes stay fixed.
- Other nodes move from place to place.
- Networks must support both static and mobile devices.



Key Takeaway



Wireless systems operate in dynamic, shared, and open environments, so they must handle mobility, interference, medium sharing, and security.

Wireless Types

Two common categories of wireless systems

1) Nomadic Systems



- ✓ Communication usually happens while the user/device is stationary.
- ✓ Often used in local wireless networks such as WLAN and WPAN.
- ✓ Can provide fast links over short range.
- ✓ Typical examples: Wi-Fi hotspot, Bluetooth devices, home/office WLAN.

 **Examples:** WLAN, WPAN, Wi-Fi, Bluetooth

2) Mobile Systems



- ✓ Communication continues while the user/device is moving.
- ✓ Designed to support mobility and handoff between cells.
- ✓ Used in wide-area cellular systems.
- ✓ Typical examples: 3G, 4G, 5G, and 6G networks.

 **Examples:** Cellular networks, mobile broadband, connected vehicles

Main Difference



Nomadic: best when the user is mostly stationary.



Mobile: best when communication must continue during movement.



Key idea: Nomadic systems emphasize fast local access, while mobile systems emphasize continuous connectivity during movement.

Wireless Networks: Infrastructure vs. Ad Hoc vs. Mesh

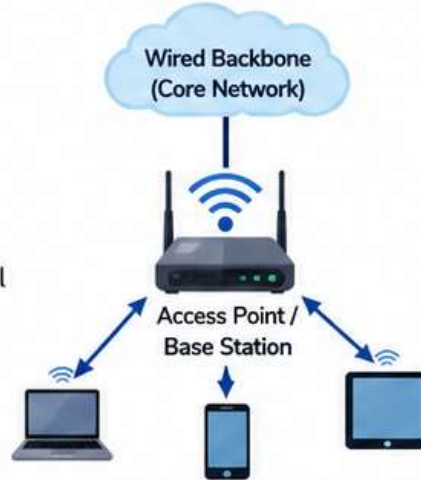
Three common wireless network architectures

1

Infrastructure-Based Wireless Networks

Uses an Access Point (AP) or Base Station (BS) as a **central node**.

- AP/BS connects wireless users to the wired backbone or core network.
- Usually follows a star topology.
- Supports centralized control and handoff.
- Common in cellular networks and standard Wi-Fi WLANs.



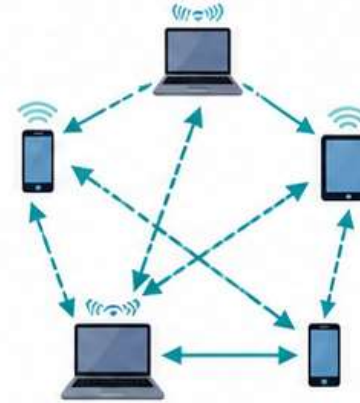
Best for: organized coverage, managed networks

2

Ad Hoc Networks

Devices communicate **directly** with one another without a fixed AP.

- Peer-to-peer topology.
- Can support multi-hop transmission.
- Flexible and quick to deploy.
- Examples: temporary networks, emergency communication, Bluetooth, 802.11 ad hoc mode.



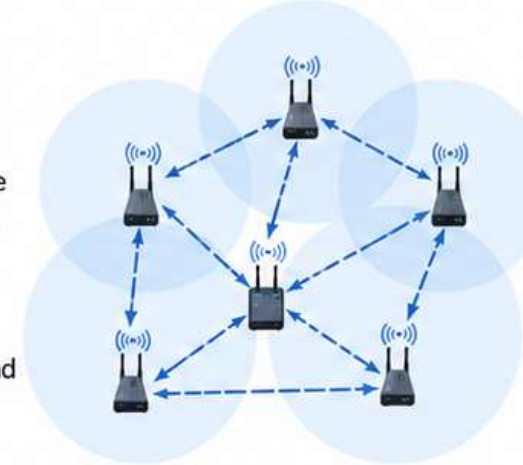
Best for: temporary, decentralized communication

3

Wireless Mesh Networks

Multiple wireless nodes cooperate to **extend network coverage**.

- Nodes relay traffic for one another.
- Provides wider coverage and improved reliability.
- Supports self-healing routes if one link fails.
- Used in campus, city, and community wireless infrastructure.



Best for: extended coverage and resilient wireless infrastructure

	Infrastructure	Ad Hoc	Mesh
 Central controller	Yes	No	Partial-distributed
 Topology	Star	Peer-to-peer	Multi-hop mesh
 Mobility / Handoff	Strong	Limited	Moderate
 Deployment style	Planned	Fast temporary	Scalable coverage
 Example use	Wi-Fi AP, Cellular	Bluetooth, emergency	Smart city, campus



Key Idea: Infrastructure networks use a **central access point**, ad hoc networks connect **peers directly**, and mesh networks extend coverage through **cooperative relay nodes**.

Elements of a Wireless Network

Main components that make wireless communication possible

1 Wireless Hosts

- End devices used by users or applications
- Can be mobile or stationary
- Send and receive data over wireless links



Smartphone



Laptop



Tablet



IoT / Sensor

3 Wireless Links



- Carry data through the air using radio signals
- May be direct or through an access point
- Quality depends on distance, interference, and obstacles

Network Infrastructure



The core network (e.g., Internet, WAN, cloud) that connects everything together.

2 Base Stations / Access Points



Wi-Fi Access Point



Cellular Base Station

- Connect wireless devices to the network infrastructure
- Coordinate communication inside a coverage area
- Examples: Wi-Fi AP, cellular base station

How they work together



User devices communicate through wireless links, often via an access point or base station, to reach the wider network.

Wireless Network Taxonomy

Classifying wireless networks by infrastructure and hop count



With
Infrastructure

Single Hop

Infrastructure Single-Hop

A wireless host connects directly to an access point or base station, which then connects to the core network or Internet.



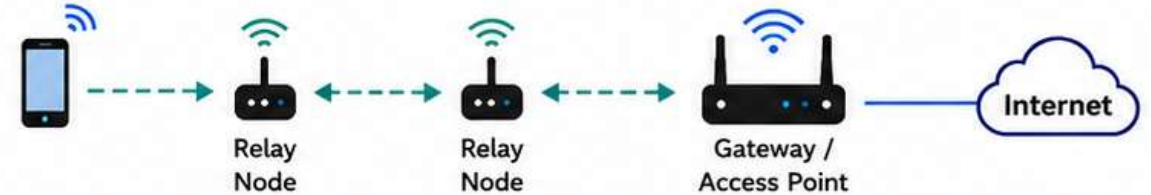
Examples

- Wi-Fi with an Access Point
- Cellular (4G/5G)
- WiMAX

Multiple Hops

Infrastructure Multi-Hop

A host reaches the network through one or more relay nodes before reaching an access point or gateway.



Examples

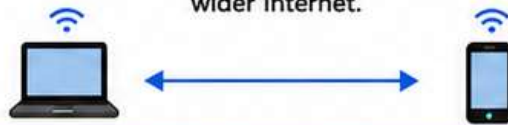
- Wireless Mesh with Gateway
- Multi-hop backhaul
- Community mesh with Internet access



Without
Infrastructure

Ad Hoc Single-Hop

Devices communicate directly with each other without an access point or connection to the wider Internet.



Examples

- Bluetooth
- Wi-Fi Direct
- Simple ad hoc links

Ad Hoc Multi-Hop

Devices forward packets for one another over several hops, without a fixed base station.



Examples

- MANET
- VANET
- Sensor multi-hop networks



Key Idea

Infrastructure networks rely on access points or base stations. **Ad hoc** networks do not. **Single-hop** means one direct wireless hop; **multiple-hop** means data is relayed through intermediate wireless nodes.

Thanks!

You've mastered the signals,
now go connect the world!



Great communication
starts with strong concepts!



Signals
Today,
Connections
Tomorrow!



Keep learning.
Keep building.
Keep connecting!



The world
needs good
communicators!

